

Name Bhavya rana

Sapid 590034689

Batch 15

Algorithm

Main Algorithm

1. Start.
2. Display the menu:
 - Remove all vowels from a string.
 - Find the first repeating lowercase alphabet in a string.
 - Check whether two strings are anagrams.
 - Exit.
3. Read the user's choice.
4. Use a **switch** statement to perform the selected operation.
5. Call a separate function for each operation.
6. Repeat the menu until the user chooses Exit.
7. Stop.

Operation 1: Remove all vowels

1. Read a string.
2. Examine each character.
3. If the character is not **a, e, i, o, u** in either uppercase or lowercase, copy it into the resulting string.
4. Terminate the resulting string with '`\0`'.
5. Display the string without vowels.

Operation 2: First repeating lowercase alphabet

1. Read a string.
2. Create a frequency array of size 26 and initialize it to zero.
3. Scan the string from left to right.
4. For each lowercase alphabet (**a** to **z**):
 - If it has already occurred, return that character.
 - Otherwise, mark it as occurring.
5. If no lowercase alphabet repeats, report that no repeating lowercase alphabet exists.

Operation 3: Check anagrams

1. Read two strings.
2. Create two frequency arrays of size 26 and initialize them to zero.
3. Scan both strings.
4. For each alphabetic character, convert it to lowercase and increment its corresponding frequency.
5. Compare the two frequency arrays.
6. If every frequency is equal, the strings are anagrams; otherwise, they are not.

C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

#define MAX 200

void removeVowels(char str[]);
char firstRepeatingLowercase(const char str[]);
int areAnagrams(const char str1[], const char str2[]);

int main()
{
    int choice;
    char str1[MAX], str2[MAX];
    char repeated;

    do
    {
        printf("\n===== MENU =====\n");
        printf("1. Remove all vowels from a string\n");
        printf("2. Find the first repeating lowercase\n");
        printf("alphabet\n");
        printf("3. Check if two strings are anagrams\n");
        printf("4. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();    // Remove newline left by scanf()

        switch (choice)
        {
            case 1:
                printf("Enter a string: ");
                fgets(str1, MAX, stdin);
                str1[strcspn(str1, "\n")] = '\0';

                removeVowels(str1);

                printf("String after removing vowels: %s\n",
str1);
                break;
```

```

        case 2:
            printf("Enter a string: ");
            fgets(str1, MAX, stdin);
            str1[strcspn(str1, "\n")] = '\0';

            repeated = firstRepeatingLowercase(str1);

            if (repeated != '\0')
                printf("First repeating lowercase
alphabet: %c\n",
                    repeated);
            else
                printf("No repeating lowercase alphabet
found.\n");

            break;

        case 3:
            printf("Enter first string: ");
            fgets(str1, MAX, stdin);
            str1[strcspn(str1, "\n")] = '\0';

            printf("Enter second string: ");
            fgets(str2, MAX, stdin);
            str2[strcspn(str2, "\n")] = '\0';

            if (areAnagrams(str1, str2))
                printf("The two strings are anagrams.
\n");
            else
                printf("The two strings are not anagrams.
\n");

            break;

        case 4:
            printf("Exiting program...\n");
            break;

        default:
            printf("Invalid choice. Please enter 1 to
4.\n");
    }

    } while (choice != 4);

```

```

    return 0;
}

/* Function to remove all vowels */
void removeVowels(char str[])
{
    int i, j = 0;

    for (i = 0; str[i] != '\0'; i++)
    {
        char ch = tolower((unsigned char)str[i]);

        if (ch != 'a' && ch != 'e' &&
            ch != 'i' && ch != 'o' && ch != 'u')
        {
            str[j] = str[i];
            j++;
        }
    }

    str[j] = '\0';
}

/* Function to find first repeating lowercase alphabet */
char firstRepeatingLowercase(const char str[])
{
    int frequency[26] = {0};
    int i;

    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] >= 'a' && str[i] <= 'z')
        {
            int index = str[i] - 'a';

            if (frequency[index] > 0)
                return str[i];

            frequency[index]++;
        }
    }
}

```

```

        return '\0';
    }

/* Function to check whether two strings are anagrams */
int areAnagrams(const char str1[], const char str2[])
{
    int freq1[26] = {0};
    int freq2[26] = {0};
    int i;

    for (i = 0; str1[i] != '\0'; i++)
    {
        if (isalpha((unsigned char)str1[i]))
        {
            char ch = tolower((unsigned char)str1[i]);
            freq1[ch - 'a']++;
        }
    }

    for (i = 0; str2[i] != '\0'; i++)
    {
        if (isalpha((unsigned char)str2[i]))
        {
            char ch = tolower((unsigned char)str2[i]);
            freq2[ch - 'a']++;
        }
    }

    for (i = 0; i < 26; i++)
    {
        if (freq1[i] != freq2[i])
            return 0;
    }

    return 1;
}

```

Test-Case Table

Test	Operation	Input	Expected Output
1	Remove vowels	Hello World	Hll Wrld
2	Remove vowels	AEIOUaeiou	Empty string
3	Remove vowels	Rhythm	Rhythm

4	First repeating lowercase	programming	r
5	First repeating lowercase	abcdef	No repeating lowercase alphabet
6	First repeating lowercase	apple	p
7	Anagram	listen, silent	Anagrams
8	Anagram	Hello,Olelh	Anagrams
9	Anagram	hello,world	Not anagrams
10	Menu validation	Choice 7	Invalid choice
11	Exit	Choice 4	Exiting program

Output

```

Last login: Fri Oct  9 20:49:29 on console
[bhavya@Bhavyas-MacBook-Air ~ % nano day42.c
[bhavya@Bhavyas-MacBook-Air ~ % clang day42.c -o day42
[bhavya@Bhavyas-MacBook-Air ~ % ./day42

===== MENU =====
1. Remove all vowels from a string
2. Find the first repeating lowercase alphabet
3. Check if two strings are anagrams
4. Exit
Enter your choice: 1
Enter a string: Hwlllo world
String after removing vowels: Hwll wrld

===== MENU =====
1. Remove all vowels from a string
2. Find the first repeating lowercase alphabet
3. Check if two strings are anagrams
4. Exit
Enter your choice: 2
Enter a string: programming
First repeating lowercase alphabet: r

===== MENU =====
1. Remove all vowels from a string
2. Find the first repeating lowercase alphabet
3. Check if two strings are anagrams
4. Exit
Enter your choice: 3
Enter first string: listen
Enter second string: silent
The two strings are anagrams.

===== MENU =====
1. Remove all vowels from a string
2. Find the first repeating lowercase alphabet
3. Check if two strings are anagrams
4. Exit
Enter your choice: 4
Exiting program...
bhavya@Bhavyas-MacBook-Air ~ % █

```